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Paleomagnetic study of Cambrian Potsdam Group sandstones, St. Lawrence Lowlands, Quebec

M. K.-Seguin, K.V. Rao and P. Arnal

Paleomagnetic Laboratory, Department of Geology, University of Laval, Quebec City, Que. G1K 7P4 (Canada)

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We report paleomagnetic results from oriented drill core samples collected at 10 sites (80 samples) from the Covey Hill and 19 sites (96 samples) from the overlying, fossiliferous Châteauguay Formations of the gently dipping Late Cambrian Potsdam Group sandstones exposed in the St. Lawrence Lowlands of Quebec. Stepwise thermal demagnetization analyses ave revealed the presence of two predominant groups of coherent magnetizations C-1 and C-2, after simple correction for bedding tilt. The C-1 group magnetization is a stable direction ($D=332°$, $I=+18°$) with unblocking temperatures (T_{UB}) between 550 and 650°C, present in the older Covey Hill Formation; this direction is probably a chemical remanence acquired during the Covey Hill diagenesis and carried predominantly by hematite. The C-2 group magnetization ($D=322°$, $I=+9°$) is present at 13 sites of the younger Châteauguay Formation; this is probably carried by magnetite and represents a penecontemporaneous, depositional DRM, characterized by T_{UB} spectra 400–550°C. We believe that C-2 is relatively younger than C-1 based on a combination of arguments such as the presence of opposite polarities, internal consistency, similarity and common occurrence of C-1 and C-2 respectively in the Covey Hill and Châteauguay members. The corresponding paleomagnetic poles C-1 (46°N, 149°E; dp, $dm=3°$, 5°) and C-2 (37°N, 156°E; dp, $dm=2°$, 5°) are not significantly different from most of the other Late Cambrian (Dresbachian-Franconian) poles derived from sediments exposed in the southern region (Texas) of the North American craton which are also believed to have been deposited during Croixian Sauk sea transgression similar to the Potsdam sandstones. Although adequate faunal control is lacking (in particular for the Covey Hill Formation), this comparison with the Cratonic poles suggests a Late Cambrian age to the Potsdam poles. The agreement between the results also gives the evidence for internal consistency of cratonic poles at least for Late Cambrian.

The incoherent C-3 group remanence ($D=250°$, $I=-15°$) is commonly present at 7 sites in both the formations; this may not correspond to a reliable paleomagnetic signal. The other remanence C-4 ($D=180°$, $I=+10°$) is found only at 3 sites located in the uppermost stratigraphic levels of the Châteauguay Formation; the corresponding paleomagnetic pole (40°N, 107°E) does not differ significantly from the Ordovician and some Late Cambrian poles. The present data are insufficient to resolve a problem in apparent polar wander for Middle and Late Cambrian time posed by the existence of high-latitude poles for some strata of Middle Cambrian age and low-latitude poles for some strata of Late Cambrian age.

1. Introduction

It is apparent from recent literature on Cambrian paleomagnetic results of North America [1–5] that the data set is not only limited in number to make Paleozoic continental restorations but also suffers from several ambiguities related to: (1) geologic age, (2) type and extent of tectonic movements of the rock units studied, (3) likely possibility of events such as Late Paleozoic or Early Mesozoic remagnetization, and (4) breakdown in the dipole nature of the geomagnetic field during Cambrian times. These factors (1 to 4) impose problems in the determination of the time of acquisition of different magnetic components. In two recent studies, Watts et al. [3,4] have favoured a Cambrian loop in the apparent polar wander (APW) path with respect to North America

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as the simplest and most unifying hypothesis to explain the distribution of Cambrian paleopoles (which spread approximately from equator, 160°E to 50°N, 100°E), in preference to other hypotheses which invoke tectonic rotations, remagnetizations and non-dipole behaviour. Paleomagnetic sampling localities for the majority of the reported studies come from the cratonic part of North America (western and southwestern parts of U.S.A.) and relatively few from Appalachians and Cordillera. In contrast to the paleomagnetic data derived from Paleozoic orogenic belts (for example from Cordilleran geosyncline in Nevada [1]), the data from platform sediments do not suffer at least a priori from the effects of major tectonic displacements and vertical axis tectonic rotations (for example from Cambrian strata of Grand Canyon in Arizona [6]) but the predominantly undeformed nature of the cratonic-cover sediments precludes the application of Graham's [7] fold test and its significance [8] to investigate the stability and infer the age of magnetization. While paleomagnetic studies made on Cambrian intrusives, for example from Wichita granites of Oklahoma [9–11], have proven to be unsuccessful in recovering the original thermoremanence, recently reported investigations from a Colorado alkalic intrusive complex [12–14] resulted in the isolation of three groups of magnetization directions which were convincingly proved to be unaffected by subsequent remagnetization.

Apart from a single study on Lower Paleozoic Bradore sandstones, western Newfoundland [15], and the two other recently reported results from Sept-Iles anorthosite [16] and Buckingham intrusives [17], no Cambrian paleomagnetic results were available from the east-central part of the craton to test the internal consistency of Cambrian paleopoles by comparison with those obtained from the southwestern side of the North American craton. The present paleomagnetic study is from the Cambrian Potsdam Group sediments located in the St. Lawrence Lowlands of Quebec. The objectives of the study are three-fold: (1) to obtain a new Cambrian result from the northern part of the craton, (2) to compare this result with those obtained from other contemporaneous sediments of the North American platform, and (3) to study the time relationships of the Cambrian Sauk sequence [18] with the help of paleomagnetic data.

2. Geologic setting

The present study area lies within the St. Lawrence Lowlands of Quebec which is a small and low-lying region of Cambrian to Ordovician flat-lying sedimentary rocks separating the Precambrian Shield on the northwest from the Appalachians on the southeast (Fig. 1). The Potsdam Group sediments of Cambrian age constitute the basal Paleozoic beds which rest unconformably on the Precambrian crystalline rocks. These sediments represent the Croixian time-transgressive series of the Early Paleozoic Sauk sequence [18] as the epeiric Sauk sea covered the eroded surface of the craton.

The geology of the Potsdam Group in southern Quebec near the Beauharnois, Ste. Anne-de-Bellevue (west of Montreal) (Fig. 1A) and the Joliette (east of Montreal) (Fig. 1B) areas is described in more detail elsewhere [19–24]. In both areas, the Potsdam Group is divisible into two formations separated from each other by a not-too well-defined unconformity; the Covey Hill Formation which rests unconformably on the Precambrian and the Châteauguay Formation at the top. The Covey Hill unit is composed of basal pebble conglomerate and heterogenerous fluviatile siltstone and sandstone with a variation in colour ranging from buff to brownish red with hematitic pigmentation and is characterized by abundant trough stratification. The formation is typified by poor sorting and irregular bedding with subrounded grains. The unconformably overlying Châteauguay Formation is mostly white to off-white well-sorted orthoquartzite with some beds of dolomite towards the top. This is characterized by its even bedding and abundant ripple marks. Structures and textures are compatible with beach and tidal environments of deposition.

The Potsdam Group is barren of fossils except for local occurrence of trace fossils abundant in some layers of the Châteauguay Formation. Inferences based on this diagnostic fossil assemblage and by the presence of Early Ordovician fossils in

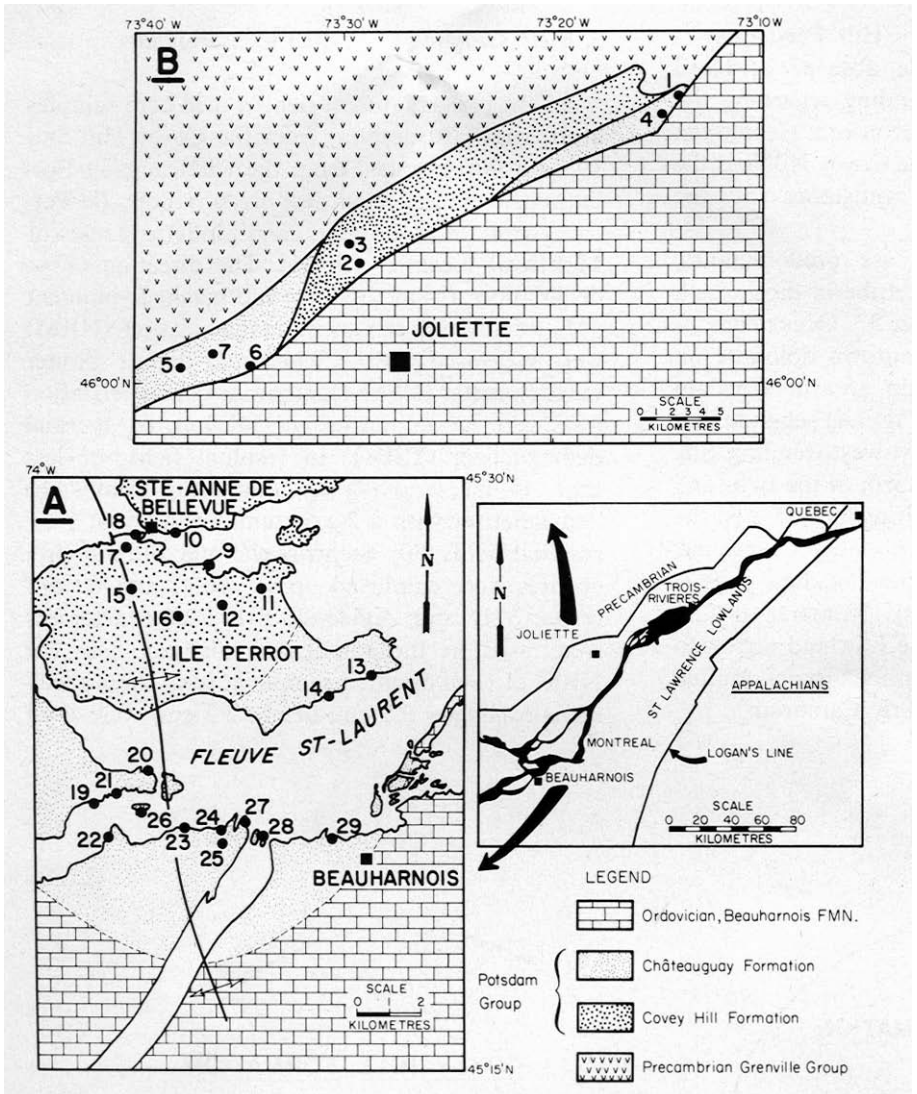


Fig. 1. Simplified geologic map of St. Lawrence Lowlands [19] and the distribution of paleomagnetic sampling sites as indicated by numbers next to solid circles in the vicinity of (A) Beauharnois and (B) Joliette areas near Montreal. The location of site 8 in the Châteauguay Formation is not shown; it is 20 km north of Ste. Anne-de-Bellevue (A).

the immediately overlying carbonates of the Beekmantown Group and with the help of stratigraphic correlations elsewhere led Clark [19] to assign a Late Cambrian age to the Châteauguay Formation. Fossils are absent in the Covey Hill Formation of Quebec and its probable equivalents in New York State. Since the nature of the unconformity between the Covey Hill and the Châteauguay Formations is interpreted as a prolonged period of non-deposition, weathering and reworking of the Lowland basin and further subsidence of this basin by a resurgence in faulting, the age of Covey Hill could be older than Châteauguay and may well be Middle or Early Cambrian [22]. However, the absence of such an unconformity in New York State based on well data [25,26] and on stratigraphic correlations [27] makes this age assignment for the Covey Hill Formation uncertain. No doubt, the

age assignment for the Covey Hill Formation is admittedly speculative in the absence of fossil control and lack of understanding regarding the presence or absence of the unconformity between the felspathic sandstones of the Covey Hill and the fossiliferous, pure quartz sandstones of the Châteauguay Formations.

The Potsdam sandstones are predominantly subhorizontal with regional southerly dips which are on average not larger than 5°. Occurrence of the Early Ordovician Beekmantown dolomite on the east and west of the main area of Potsdam exposures near Beauharnois (Fig. 1A) suggests that it is gently anticlinal. An east-west-trending Ste. Anne-de-Bellevue fault runs north of the two sampling areas. While normal faulting has affected the Lowlands in Cambrian and possibly Cretaceous times, the gentle folding of the Potsdam beds is believed to be primary [28]. Kumarapeli [29] attributes the formation of the Lowland region to the development of a half-graben block-faulted structure sometimes during Early Cambrian.

3. Field sampling and NRM measurements

In the present study oriented drill core samples at 10 sites (80 samples) from the Covey Hill and 19 sites (96 samples) from the Châteauguay Formations were collected near Beauharnois, Ile Perrot (west of Montreal) and Joliette (east of Montreal) localities (Fig. 1). The direction (D = declination; I = inclination) and magnetic moment (M) of natural remanent magnetization (NRM) were measured with a Schonstedt digital spinner magnetometer (DSM-1); stepwise demagnetization tests were carried out using (1) Schonstedt thermal demagnetizer (TSD-1) in residual fields of less than 15 nT, and (2) an alternating field (AF) demagnetizer with a 3-axis tumbler in about 5 nT residual fields [30]. Stepwise thermal and AF procedures were employed up to 700°C and 80 mT respectively and Zijderveld [31] vector diagrams were used in the subsequent analysis. For the NRM of each oriented sample at least three specimens (diameter 2.5 cm, height 2.2 cm) were used

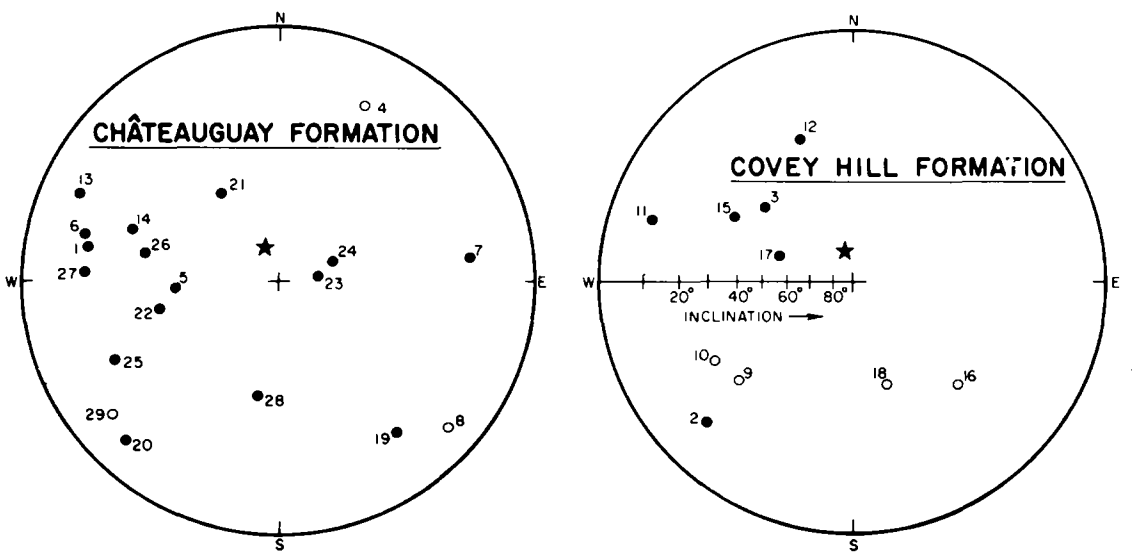


Fig. 2. The site-mean NRM directions (after tilt correction) plotted on a Wulff projection for the Châteauguay and Covey Hill Formations in the Potsdam Group sandstones. The star represents the direction of the present Earth's field at the sampling sites. Solid (open) circles represent projections on lower (upper) hemisphere.

to obtain a sample-mean direction. The sandstone samples were subjected to isothermal remanence acquisition (IRM) tests at room temperature up to about 1.5 T to infer the magnetic mineralogy.

The site-mean directions for the Covey Hill and Châteauguay Formations are corrected for the gentle tilt. These directions are widely scattered and the majority of them are far displaced from the present earth's field (PEF, $D = 343°, I = +74°$) indicating the presence of both low- and high-coercivity components of magnetization (Fig. 2). The intensities of NRM (J_n) range from 10^{-2} to 10^{-3} A m^{-1}. Single measurements of NRM were usually repeatable to within 3% in intensity and within 3° in direction.

4. AF and thermal results

4.1. Châteauguay Formation

One specimen each from 50 samples (14 sites) was demagnetized in alternating fields up to 80 mT. Most samples lost 50% of their NRM intensity by 10 mT and 90% by 50 mT, while some samples were not cleaned beyond 40 mT because of unstable magnetization during treatment. Because of weak remanence associated with the samples, care was taken to minimize the effects of spurious magnetizations introduced in the demagnetization process beyond 40 mT. Most of the samples have a straight decay to the origin of the magnetization vector above 10 mT which may mean removal of only one component of magnetization. The hard component of some samples (sites 4, 7, 8 and 19) is consistently southeast with a shallow inclination, in contrast to the most commonly observed antipolar direction with northwest declinations and shallow inclinations in the majority of samples from the other nine sites (5, 6, 13, 14, 20, 21, 25, 26 and 27). This may imply the presence of both normal and reverse polarity zones in the Châteauguay Formation. Unfortunately, the Potsdam sandstone in the Lowland region is nowhere exposed in its entirety; most exposures are thin and discontinuous. No marker horizons are known and these outcrops are generally located at unknown stratigraphic elevation above the Precambrian. The non-availability of stratigraphically continuous sections in the sampling areas precludes positioning of the paleomagnetic sites with respect to each other to locate the boundary between the two polarity zones. Because of incomplete cleaning of some samples at higher fields (> 40 mT) due to the presence of spurious magnetizations, thermal treatment was preferred to obtain a clear decay to the origin.

Two samples form each of the 19 Châteauguay sites were subjected to thermal treatment in 100°C steps up to 400°C and in closer temperature steps up to 600°C until they showed a straight vector decay to the origin. Two representative demagnetization trajectories for specimens 13B2 and 26L are shown in Fig. 3. These show that an initial rema-

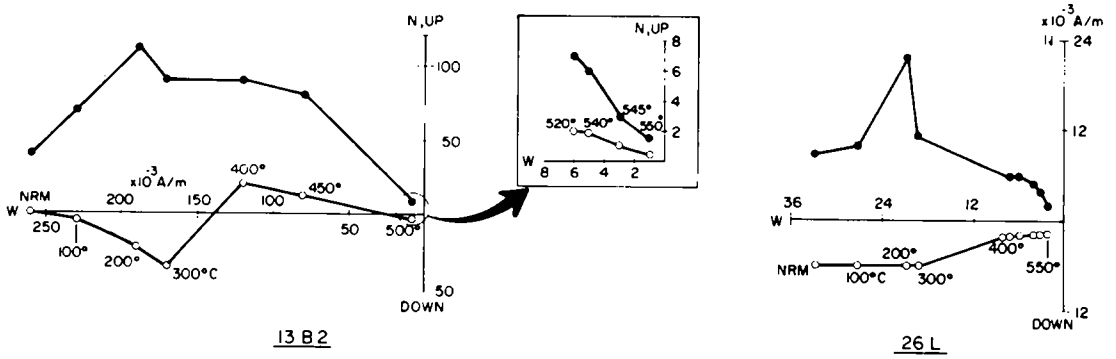


Fig. 3. Representative orthogonal thermal demagnetization diagrams [31] of successive end-points of magnetization vector for two Châteauguay specimens as indicated. Solid (open) circles represent projections on horizontal (vertical) plane. Numbers next to open circles represent the demagnetization steps in °C. Units indicated for axes are in 10^{-3} A m^{-1} ($= 10^{-6}$ emu cm^{-3}).

nence with $D = 290°$, $I = +10°$ is removed below 400°C and an intensity of 35% (26L) to 65% (13B2) of NRM. Above 400°C and mostly between the 500 and 550°C steps, a straight decay of the magnetization vector was observed with a direction $D = 320°$, $I = +10°$. This high-temperature direction is not significantly different from the initial direction (except for a 10–30° difference in D) and is present in 70° of samples; this direction is also in agreement with the AF demagnetization results obtained from companion specimens with coercivity spectra of 30–50 mT. A single southeast direction represented by a linear vectorial decay to the origin, antiparallel to the above northwesterly remanence, is also removed in thermal treatment between 500 and 550°C steps from sites, 4, 7, 8 and 19. However, the samples from the three sites 22 ($D = 251°$, $I = -8°$), 23 ($D = 71°$, $I = +8°$) and 24 ($D = 68°$, $I = +16°$) have shown different directions over a wide range of temperatures but in association with no other remanence. These are not well-defined by Zijderveld trajectories relative to the northwest and southeast directions. At three other sites (1, 28 and 29) a southerly directed shallow remanence ($D = 180°$, $I = +10°$) is observed. The collected samples at these sites are dolomitic in composition and they were found to be very weakly magnetized after thermal treatment beyond 500°C.

4.2. *Covey Hill Formation*

Pilot AF demagnetization of the majority of samples from the Covey Hill collection showed no significant changes from their respective NRM directions or intensity up to 80 mT, while some of them have shown a very erratic behaviour during demagnetization runs.

Since AF treatment did not prove to be very effective, the Covey Hill samples were thermally demagnetized in steps of 100°C up to 500°C beyond which treatment was carried out in closer temperature steps up to 700°C. Representative vector diagrams of thermal demagnetization for

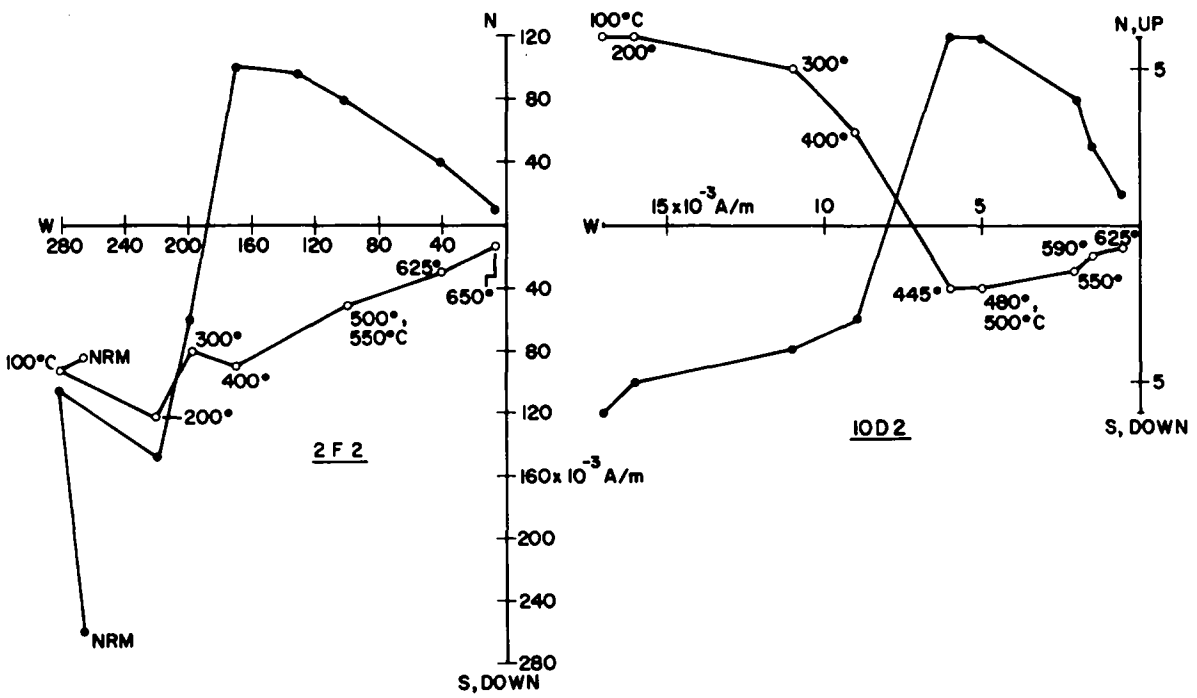


Fig. 4. Representative orthogonal thermal demagnetization diagrams of successive end-points of magnetization vector for two Covey Hill specimens as indicated. The other symbols as in Fig. 3.

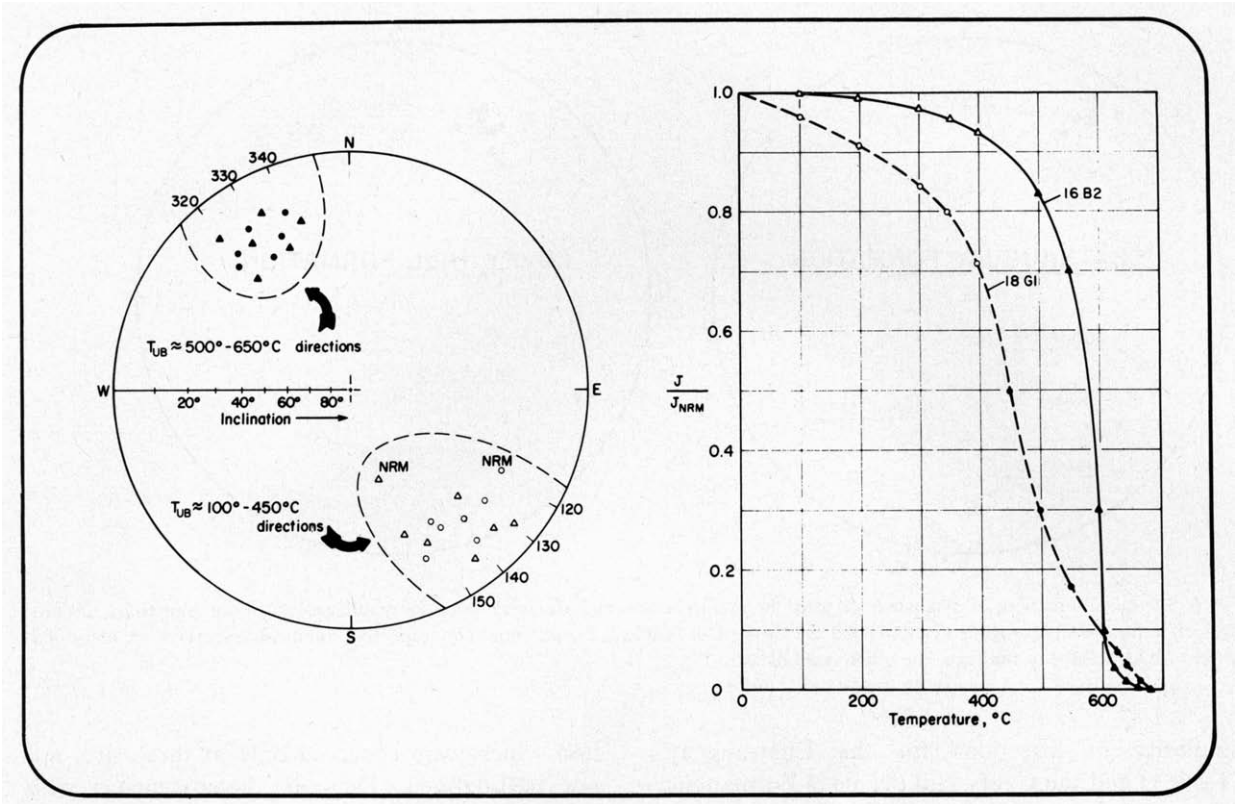


Fig. 5. Successive end points of magnetization vector during thermal demagnetization plotted on a Wulff projection (left) and the corresponding normalized thermal decay curves (right) for two Covey Hill specimens. The other symbols as in Fig. 2.

two specimens (2F2, 10D2) are shown in Fig. 4. They show the presence of a stable and dominant northwesterly directed remanence at temperatures as high as 650°C but also accompanied by a less common southwesterly remanence with poorly defined T_{UB} spectra. The low T_{UB} direction ($\leq$ 400°C) is not well-defined and is present in sixteen samples only. The presence of antiparallel directions associated respectively with low (100–450°C) and high (500–650°C) unblocking temperature (T_{UB}) spectra are present at sites 2, 16 and 18 as shown by two specimens 16B2 and 18G1 (Fig. 5). The normalized thermal decay curves show discrete (16B2) and distributive (18G1) types [32] which suggest the presence of hematite as the most probable and dominant carrier of remanence. However, the presence of antiparallel directions during thermal treatment is not a general phenomenon.

5. Analysis of paleomagnetic directions

Since AF treatment did not prove to be completely effective in the resolution of different magnetization directions, demagnetization results obtained from thermal demagnetization are preferred. The rest of the collection in both the Châteauguay and Covey Hill Formations was thermally treated and statistical analysis is based only on these results. All the site mean directions are corrected for the gentle geological tilt but without resulting in a significant improvement relative to the in-situ dispersion values of directions. The various components of magnetization were resolved by examining Zijderveld plots for linear demagnetization trajectories defined by three or more points. The site-mean directions of magnetization were computed and classified into four groups (C-1, C-2, C-3 and C-4) based on the

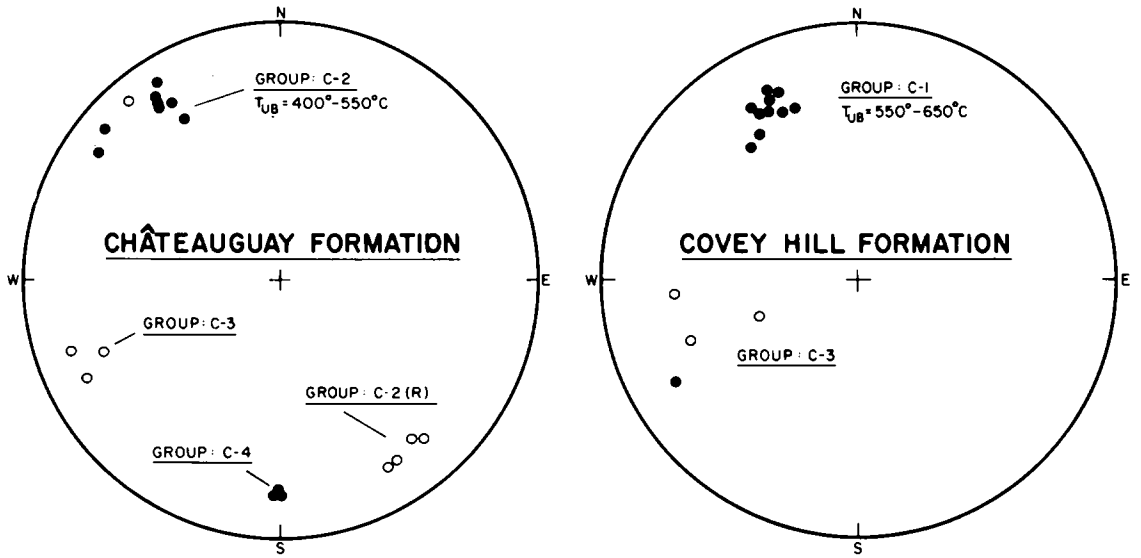


Fig. 6. Site-mean directions of magnetization after thermal treatment and after correction for gentle geological dip, plotted on a Wulff projection for the Châteauguay (Table 1) and the Covey Hill (Table 2) Formations. The respective site-mean directions are identified groupwise as defined in the text. The other symbols as in Fig. 2.

similarity in directions for the Châteauguay (Table 1) and the Covey Hill (Table 2) Formations (Fig. 6).

5.1. Châteauguay Formation

Based on the criteria such as stable end-points, range of T_{UB} spectra and the magnitude of within-site scatter, three groups of magnetization directions are recognized, in the decreasing order of stability from the most stable C-2, intermediate C-4 and to the least stable C-3 group directions in the Châteauguay sediments (Table 1). The C-2 group is a northwesterly directed remanence with shallow, positive (downward) inclination. It is univectorial and magnetically hard. It characterizes the Châteauguay Formation because it is observed at the majority of the sites and obtained from samples with stable end-points. A group of directions approximately antipodal to C-2 (denoted as C-2 (R) in Table 1) but with similar stability characteristics is also found at four sites. As mentioned earlier, it is not known with certainty whether this indeed represents the presence of a field reversal during the deposition of Châteauguay sediments. In contrast, the characteristics of C-3 magnetization which were observed only at three sites are not well-defined. They are heterogeneous with considerable scatter from sample to sample as well as large within and between site scatter and variable T_{UB} and coercivity spectra. However, this direction cannot be a soft viscous magnetization caused by PEF because its inclinations are too shallow. The intermediate C-4 group remanence directions are typified by southerly declinations and shallow positive inclinations. The remanence is univectorial and present only at three sites 1, 28 and 29 which happen to occupy the uppermost dolomitic beds of the Châteauguay Formation. The location of the respective sites in Fig. 1 suggests that they stratigraphically border the overlying Early Ordovician Beauharnois Formation.

Moderate to large increases in the within-site dispersion and an order of magnitude increase in the between-site dispersion relative to NRM values (not shown) for groups C-2, C-2 (R) and C-4 suggest that the distinction between the remanences is not mineralogical but concerns possibly differences in the mode and time of acquisition of each group. Tests based on stability criteria such as Graham's [7] fold test cannot be performed to infer the age of magnetization of each group with

TABLE 1

Site-mean remanence directions after thermal treatment * —Châteauguay Formation

Site No.	N	D(°)	I (°)	K	α_{95} (°)
Group C-2					
5	7	329	+17	25	12.5
6	5	325	+11	64	9.6
13	4	319	−5	144	8.0
14	3	326	−8	191	9.0
20	7	328	+5	21	13.5
21	2	329	+12	–	–
25	5	305	+8	198	6.0
26	2	323	+10	–	–
27	4	310	+5	61	11.9
Group C-2 (R)					
4	7	140	−12	46	9.0
7	4	130	−9	38	15.0
8	4	147	−9	83	10.1
19	6	138	+10	95	7.0
Group C-3					
22	5	251	−8	26	15.3
23	3	243	−9	11	39.0
24	8	248	−16	7	22.0
Group C-4					
1	3	181	+10	325	6.9
28	4	179	+11	52	12.9
29	4	180	+10	62	11.8

N= number of samples used; D, I=declination and inclination of remanent magnetization in degrees, K=estimate of Fisher's precision parameter; α_{95}=semi-angle of cone of 95% confidence for mean direction.

* Approximate range of unblocking temperatures, T_{UB}, over which mean directions are calculated, respectively for groups C-2 and C-2 (R), C-3 and C-4 are 400–550°, 100–500°C, and 400–550°C.

respect to the time of deformation in view of the gentle and almost uniform geological dips observed at the sites. It is very plausible that the C-2 and C-2 (R) remanences at Châteauguay sites were acquired earlier than the C-4 remanence which is common only at three sites which occupy the uppermost section of the Châteauguay Formation. Relative to C-2 and C-4, the time of acquisition of C-3 cannot be ascertained except for the fact that it was observed as a secondary remanence to C-1 group directions in the underlying Covey Hill sandstones.

5.2. *Covey Hill Formation*

The stable magnetization with coherent directions observed in all the Covey Hill samples is predominantly of C-1 group which is univectorial. This direction also occurs at four sites in association with an ill-defined low T_{UB}, C-3 group directions (Table 2). The C-1 group remanence is a northwesterly directed magnetization similar to C-2 in stability characteristics but it resides only in a high T_{UB} spectra (550–650°C) and differs by at least 100°C from that of C-2 spectra (400–550°C). At three sites (2, 16 and 18), the samples showed opposite polarity directions with southeasterly and northwesterly declinations associated respectively with low (100–450°C) and high (500–650°) T_{UB} spectra (Fig. 5). The presence of approximately antiparallel directions in the same sample may mean that the magnetizations of both the polarities were acquired geologically in a short interval of time and that the magnetization could perhaps be a chemical remanence acquired by oxidation of detrital magnetite to hematite. The southeasterly

TABLE 2

Site-mean remanence directions after thermal treatment *—Covey Hill Formation

Site No.	N	D (°)	I (°)	K	α_{95} (°)
Group C-1					
2	5	332	+17 (R)	91	8.1
3	2	340	+29		
9	4	321	+23	252	5.8
10	6	335	+11	31	12.3
11	4	336	+19	61	11.9
12	3	337	+13	157	10.0
15	6	330	+16	58	8.9
16	7	328	+14 (R)	45	9.1
17	4	326	+22	164	7.0
18	5	334	+14 (R)	50	10.9
Group C-3					
2	3	241	+12	29	23.4
9	6	249	−45	9	23.2
10	3	250	−20	51	17.5
12	4	266	−19	13	21.8

Notation of symbols as in Table 1.

* Approximate range of unblocking temperatures, T_{UB}, over which mean directions are calculated respectively for groups C-1 and C-3 are 550–650°C and 100–500°C.

directed, low T_{UB} remanence is similar to C-2 (R)-type (Table 1) directions observed in the Châteauguay sites 4, 7, 8 and 19. It is difficult to establish whether this similarity in southeasterly directions is fortuitous but the C-1 group directions in the Covey Hill are also accompanied by heterogeneous C-3 group remanence. The relative ages of C-1 and C-3 magnetizations and the similarity in the southeasterly directions of C-1 in the Covey Hill sites with the C-2 (R) directions of the Châteauguay Formation cannot be ascertained in view of the gently dipping strata.

6. Magnetic mineralogy

The C-1 and C-2 group remanences (either with northwest or southeasterly declinations) which occur respectively at majority of the sites in the Covey Hill and the Châteauguay formations are almost similar in directions and stability characteristics but differ only in the T_{UB} spectra in which they reside. In order to determine the magnetic mineralogy of the Potsdam samples, we measured the build-up of isothermal remanence (IRM) and the remanence-coercivity (H_{cr}) at room temperature in fields up to 1.4 T. Since in sedimentary rocks magnetite, black specularite (> 1 μm) and red pigment (< 1 μm) cover roughly the ranges < 0.1 T, 0.1–0.3 T and > 0.3 T, respectively, of the H_{cr} spectrum, the IRM acquisition method is very useful [33], particularly for weakly magnetic rocks, for detecting the magnetic minerals present. Fourteen representative specimens (8 from Châteauguay, 6 from Covey Hill) were selected for the experiments but build-up curves are shown only for four specimens (Fig. 7). The two specimens (5G2, 25B2) from the Châteauguay Formation reach saturation within an applied field of 0.1 T and the values of $H_{cr} = 0.03$–0.05 T (as noted on the back field curves) indicating the presence of magnetite. These two specimens carry univectorial group C-2 remanence (Table 1) typical of many Châteauguay sites. The other specimens (20B1 with C-2; 19G2 with C-2 (R); 22L, 24L1 with C-3; 28I, 29G2 with C-4) were also found to contain magnetite identified by low-field (< 0.1 T) saturation and H_{cr} values 0.02–0.05 T. These results are in accordance with intermediate coercivity (15–50 mT) and intermediate T_{UB} (200–550°C) spectra in

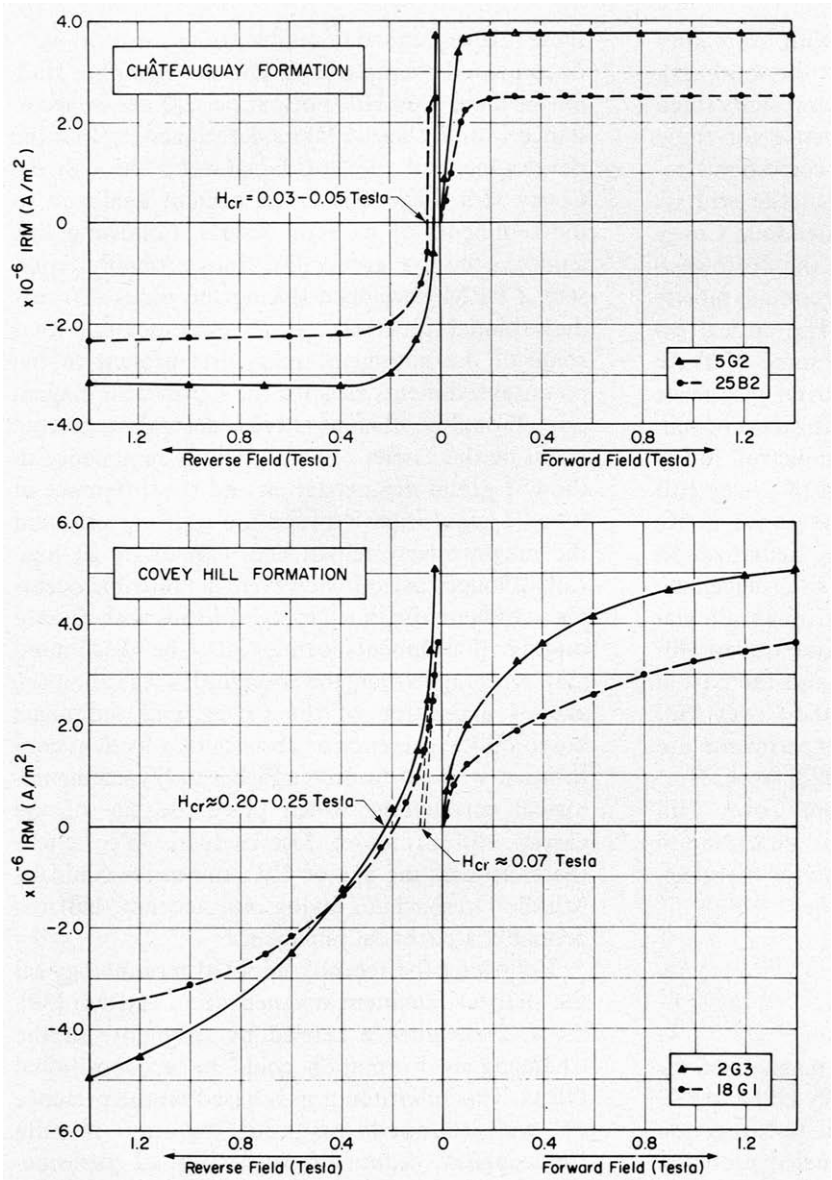


Fig. 7. Acquisition of isothermal remanence (A m^{-2}) at room temperature, in forward and reverse field increments up to 1.4 T for two specimens (as indicated) each from the Châteauguay (top) and the Covey Hill (bottom) Formations. The intersections of curves in the reverse field axis corresponds to the values of coercivity of remanence (H_{cr}) as indicated.

AF and thermal treatments obtained for C-2, C-3 and C-4 remanences. This is confirmed by the optical examination of a few polished sections for these samples which showed the presence of magnetite with high-sphericity indices indicating that their deposition was penecontemporaneous with the deposition of the Châteaugauy sediments.

In contrast the acquisition curves for the two Covey Hill specimens (2G3 and 18G1) did not reach saturation even at fields as high as 1.4 T, which is typical of hematite in either specular or red pigment form; also, the corresponding backfield curves suggest the presence of hematite with high values of H_{cr} = 0.20–0.25 T and also mag-

netite (based on sharp inflections) with low values of H_{cr} (0.07 T). This may mean that the southeasterly (intermediate T_{UB}) and the northwesterly (high T_{UB}) directions of the C-1 remanence for these two specimens (Fig. 5 for 18G1) correspond respectively to those of detrital magnetite and its oxidized mineral hematite. The other four Covey Hill specimens which carried only the northwesterly C-1 remanence were inferred to contain mostly hematite from IRM experiments. The optical examination of polished sections of some of these samples reveal the presence of detrital magnetite with partly hematized grains. Hematization of biotite flakes characterized by submetallic red flakes is also very common in thin sections of Covey Hill samples. Lewis [28] proposed a qualitative petrographic interpretation regarding the hematization of biotite flakes in terms of in situ diagenetic processes during the accumulation of an alluvial complex supplied by eolian action at Covey Hill times. The occurrence of hematite and the partial presence of detrital magnetite in the Covey Hill samples as revealed by the IRM experiments are in excellent agreement with Lewis' [28] work. Normalized thermal decay curves from Covey Hill samples also point to possible small contribution from magnetite (thermal decay curve for the specimen 18G1, Fig. 5) in the T_{UB} spectra.

7. Age and origin of magnetizations

For the majority of the sites the magnetizations were predominantly C-1 in the Covey Hill and C-2 in the Châteauguay Formations. The C-1 direction is a stable, high T_{UB} remanence carried probably by hematite as evidenced by the AF and thermal behaviour, IRM results and optical studies. The origin of hematite in the desert-originated sandstones of the Covey Hill Formation could perhaps be attributed to the same intrastratal alteration processes responsible for the generation of hematite as proposed for the Tertiary red beds of Sonoran desert, Baja California [34–36]. According to Walker [34] the growth of hematite in such conditions may take 10^4 to 10^6 years. Development of chemical remanent magnetization (CRM) during the early stages of red bed formation in these Late Cenozoic sediments is also a well-documented example [37]. The petrographic studies on the Covey Hill Formation [28] are in accordance with Walker's [34] model suggesting that the development of the Potsdam Group through the Covey Hill diagenesis as an ancient analogue to the sediments of modern deserts. Following this analogy, the C-1 group directions probably represent a CRM developed during the times of post-depositional processes; it is quite possible that some of the authigenic magnetite present in the parental sediments escaped the Covey Hill diagenesis. Probably, this unaltered detrital magnetite could be the carrier of southeasterly remanence in the C-1 group magnetization and the difference in time of acquisition between the southeasterly and the northwesterly remanences cannot be geologically a longer period. However, the possible occurrence of hematite in a variety of forms with diverse origins in sediments cannot also be discounted [38]. Keeping in view the ambiguities regarding the age of deposition of the Covey Hill sediments based on the presence or absence of a local unconformity, we tend to prefer Fisher's [27] sedimentological correlations which place the age of the Covey Hill at about Dresbachian. We believe therefore that the age of C-1 remanence could be Middle Dresbachian taking into account that it is probably a chemical remanence.

Following the recently adopted terminology on the detrital remanent magnetization (DRM) [39], the C-2 remanence carried by magnetite in the Châteauguay Formation could be a depositional DRM. This interpretation is based on the presence of well-rounded magnetite grains in the Châteauguay sediments suggestive of penecontemporaneous deposition. The greater sphericity of sands and opaques and relative lack of strained and polycrystalline quartz in the Châteauguay Formation indicates that it was subjected to a longer and greater abrasion history than the Covey Hill sands. According to Lewis [28], the Châteauguay Formation fits a classic picture of a quartz sandstone-carbonate sequence transgressing on to a stable or slowly subsiding cratonic edge. The presence of diagnostic Late Cambrian fossils in the Châteauguay Formation indicates that its deposition occurred under marine environment,

and reworking of the uppermost Covey Hill deposits probably took place during the transgression of marine waters. The agreement in the southeasterly directions between the C-2 and C-1 remanences could be attributed to a possibility that the lower units of the Châteauguay (Cairnside Member) are the modified form of upper Covey Hill sands [28]. But this is highly unlikely since the amount of time necessary for subaerial processes must be very long to eliminate sand-size felspars and polycrystalline quartz from the Covey Hill sediments. Furthermore this time difference (= unconformity?) depends also on the then prevalent climatic conditions.

We favour that C-1 is a relatively older remanence (Mid-Dresbachian) compared to C-2 and that both differ in age at least by the duration of time necessary for subaerial processes to wipe out felspars from the Covey Hill. It is likely that the southeasterly and northwesterly remanences of C-2 group were acquired respectively one after the other before and after a field reversal. We believe therefore that the C-2 group remanence is acquired some time after Mid-Dresbachian.

The C-4 remanence is found only at three Châteauguay sites (1, 28 and 29) where the predominant rock type is dolomitic quartz sandstone. These rocks are weakly magnetic beyond the 500°C thermal treatment. The respective sites occupy the uppermost section of the Châteauguay Formation (Theresa Member) or they may well be part of Early Ordovician carbonate of the Beauharnois Formation (Fig. 1). Based on the stratigraphic location of the sites, we believe that the C-4 remanence is the youngest compared to other two remanences (C-1, C-2) and may correspond to uppermost Cambrian to early Ordovician in age.

The rather ill-defined C-3 remanence is commonly present in both the Châteauguay (at 3 sites as a univectorial direction) and the Covey Hill (at 4 sites as a low-T_{UB} component accompanying C-1) Formations. This southwesterly remanence with shallow inclinations appears to be either: (1) a vector sum of northwesterly C-2 remanence of Late Cambrian age and southerly directed C-4 remanence of Cambro-Ordovician age, or (2) following Larson and Walker [37] it represents a weaker resultant remanence that exists in some laminations of the strata which contain two or more authigenic minerals magnetized in different directions. In the absence of strong evidence for Ordovician remagnetization, we favour the second alternative as a possibility in the Potsdam sandstones and therefore, we believe that the C-3 remanence does not bear any correlation with the geomagnetic field at the time of deposition.

8. Paleomagnetic poles

Based on the group-mean remanence directions, paleomagnetic poles are calculated for each group C-1, C-2 and C-4 (Table 3). The available Cambrian and Ordovician poles for North America are compared with the Cambrian poles obtained in the present study (Fig. 8). Because the Potsdam poles (in particular C-1 from unfossiliferous Covey Hill Formation) are uncontrolled temporally from faunal evidence, a possible age might be inferred from a comparison with a spectrum of Cambrian poles of known age. A Late Cambrian age to the poles C-1 and C-2 is reasonable since they differ significantly from poles TP (Lower Cambrian, Tapeats sandstone) and MF, AF (Middle Cambrian, Muav and Abrigo Formations) obtained from stratigraphically well-studied Grand-Canyon strata of Arizona [6]. But the majority of the other Lower and Middle Cambrian poles seem to have more uncertainties in terms of age of magnetization, time of deformation and tectonism than Late Cambrian poles. Therefore, we limit our discussion to Late Cambrian poles only.

The reported Late Cambrain poles come mostly from the flat-lying, stratigraphically ordered sequence of sediments exposed in the Llano Uplift (poles HR, CR, LR, MW and PW) of Texas [3,49] and the Lamotte white sandstone (LF-1, LF-2) [42] and the overlying Bonneterre dolomite (BD) [41] of Missouri. Similar to the Potsdam sandstones of the present study, the sedimentary units from Texas and Missouri also resulted from the Sauk sea transgression [18]. Except for the Bonneterre dolomite, the rest of the sediments were found to possess high-T_{UB} magnetization directions carried by hematite. The poles C-1 and C-2 are not far displaced from the approximately con-

TABLE 3

Group-mean remanence directions and paleomagnetic poles, Potsdam sandstones, Quebec

Formation	Group * and pole **	T_{UB} ***	N	Mean magnetization				Paleopole			
				D (°)	I (°)	K	α_{95} (°)	lat.	long.	dp	dm
Châteauguay	C-4	400–550	3	180	+10	4920	1.8	40°N	107°E	1	2
	C-2, C-2 (R)	400–550	13	322	+9	78	4.7	37°N	156°E	2	5
Covey Hill	C-1	550–650	10	332	+18	111	4.6	46°N	149°E	3	5
Châteauguay and Covey Hill	C-3	100–500	7	250	−15	19	14.2	–	–	–	–

* Remanence directions as defined in Tables 1, 2 (see text for details).

** Pole symbol used in Fig. 8.

*** Approximate range of unblocking temperatures, T_{UB}, over which mean remanences are represented by linear vector trajectories. N = number of sites used in averaging; dp, dm = semi-axes of ovals of 95% confidence. Other symbols as in Table 1.

temporaneous poles HR, CR, LR and MW (upper Middle Albertan to Middle Franconian) except for the youngest pole PW (Upper Franconian). Upon consideration of errors in the respective site-mean pole determinations (C-1, C-2; HR, CR, LR, MW) and the short (~20–30 m.y.) time-span of the Croixian series (525–500 m.y. [40]) during which these sediments were mostly deposited, such an

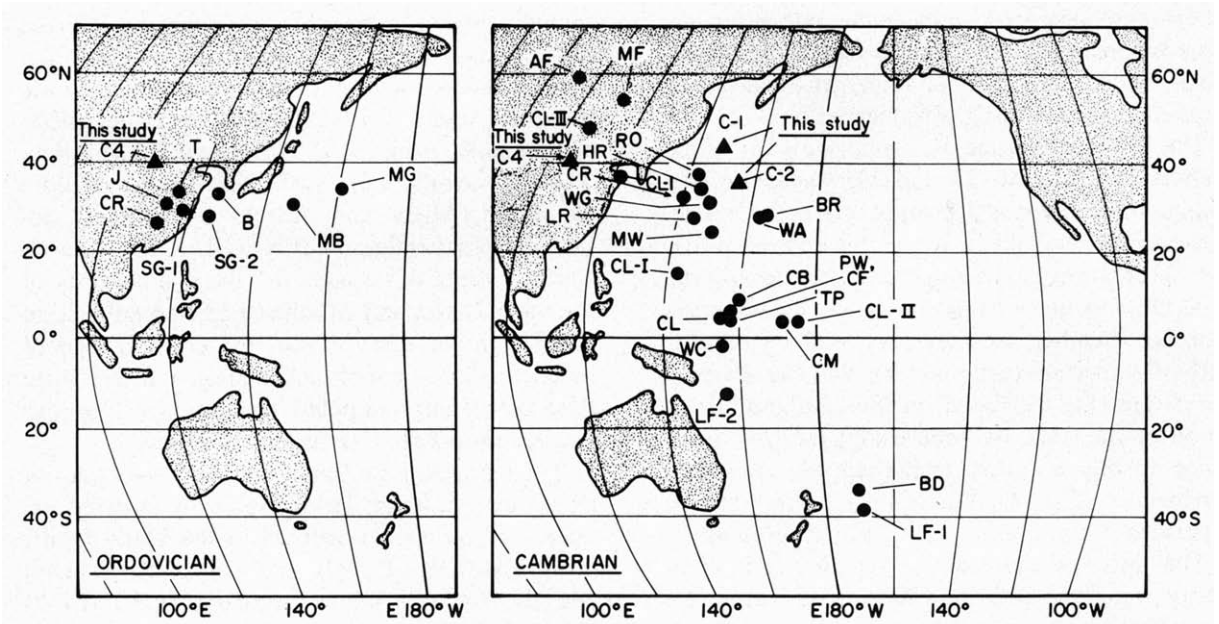


Fig. 8. Ordovician (left) and Cambrian (right) paleomagnetic poles from North America (in solid circles). Contemporaneous poles derived from (1) tectonically exotic terranes such as Avalon zone and Piedmont province, (2) transported ophiolite suites, and (3) those reported in abstract form are not included. Ordovician poles: CR = Chapman Ridge Fm [43]; MB = Moccasin and Bays Fm [43]; T = Trenton Lst [44]; B = Beemerville complex [45]; J = Juniata Fm [46]; SG-1 = St. George Lst [47]; SG-2, St. George Lst [41]; MG = Martinsburg Fm [48]. Cambrian poles: HR = Hickory, Riley Fm [3]; CR = Cap Mtn, Riley Fm [3]; LR = Lion Mtn, Riley Fm [3]; MW = Morgan-Welge, Wilberns Fm [3]; PW = Point Peak, Wilberns Fm [49]; AF = Abrigo Fm [6]; MF = Muav Fm [6]; TP = Tapeats Fm[6]; RO = Rome Fm [4]; WA = Waynesboro Fm [4]; BR = Bradore Fm [15]; CM = Cloud Mtn flows [47]; CL-I, CL-II, CL-III = Colorado intrusives [13]; CL-I = Colorado intrusive [12]; LF-1, LF-2 = Lamotte Fm [42]; BD = Bonneterre Dst [41]; CL = Caldwell Lavas [50]; CB = Carara-Bonanza [1]; CF = Carara Fm [1]; WC = Wood Canyon Fm [1]; CB, CF, WC are corrected for rotation [1]; WG = Wichita granites [11]. Poles C-1, C-2 and C-4 obtained in the present study are shown in solid triangles.

agreement between the poles derived from geographically distant parts of the North American craton is reasonable. Furthermore, this suggests the internal consistency of the cratonic poles at least during this part of the Cambrian period. It is also interesting to note that the pole WG derived from Wichita granites (~ 525 m.y. [11]) is in agreement with the poles from Texas. In contrast, the poles (not shown in Fig. 8) of approximately similar age (Middle Dresbachian) but derived from thrust belts (Tennesse, Nevada, Utah, Alberta) diverge from the Cratonic poles of Texas and the divergence was explained in terms of vertical axis tectonic rotations amounting to 15–25° [5].

The disagreement between C-1, C-2 and the other well-determined pole PW [49] by about 35° of arc may be taken as an evidence that the pole C-2 could be older than Upper Franconian but such a reasoning cannot be valid in the absence of sufficient knowledge on the relative positioning of sites occupied in the Châteauguay Formation. But the other Late Cambrian poles LF-1, LF-2 and BD are far displaced from the contemporaneous poles C-1, C-2 and also the poles from Llano Uplift. The presence of multivectorial NRM in the Lamotte sandstones and their behaviour during thermal treatment led Van der Voo et al. [49] to question the reliability of Late Cambrian poles obtained by Al-Khafaji and Vincenz [42]. With the presently available data we cannot explain the divergence between the two groups of cratonic poles LF-1, LF-2, BD of Missouri and the contemporaneous Late Cambrian poles from nearby Texas. However, one of the Lamotte poles LF-2 falls south (by about 20° in latitude) of the pole PW, i.e. on return track of the proposed Cambrian Loop [3]. The present data are insufficient to resolve the problem [3,49] posed by the presence of low latitude poles for some strata of Late Cambrian age (for example poles PW, LF-2) and the existence of high latitude poles (for example poles MF, AF) for some strata of Middle Cambrian age [6].

The pole C-4 derived from the uppermost stratigraphic sections of the Châteauguay Formation is not significantly different from the poles derived from Colorado intrusive (CL-1 [12], CL-III [13]). However, the general position of these poles is also not much different from Ordovician poles (Fig. 8). We consider therefore that the age of C-4 magnetization most likely corresponds to uppermost Cambrian/Early Ordovician. The possibility of Early Ordovician C-4 magnetization could be a consequence of the slow shift of both the epeiric seashore line and the boundary between limestone and sandstone facies resulting in an entirely Late Cambrian sequence in New York (south) and partly Early Ordovician in Quebec (north).

9. Summary and conclusions

A paleomagnetic study of the Potsdam Group sandstone exposed in the St. Lawrence Lowlands of Quebec revealed the presence of two predominant groups of coherent magnetizations C-1 and C-2. The C-1 group remanence is a stable, high-T_{UB} (550–650°C) direction ($D = 332°$, $I = +18°$) present at 10 sites of the Covey Hill Formation. This direction is probably acquired by CRM and carried predominantly by hematite developed during the Covey Hill diagenesis. The C-2 group remanence ($D = 322°$, $I = +9°$) is obtained with stable end-points and characterized by T_{UB} spectra, 400–550°C, at 13 sites of the Châteauguay Formation; this magnetization is probably carried by magnetite and represents a depositional DRM acquired penecontemporaneous to the deposition of Châteauguay sediments. Unlike the C-1 and C-2 groups, the C-3 group remanence ($D = 250°$, $I = -15°$) is heterogeneous and always associated with a large within and between site scatter, but commonly present in both formations. The incoherent C-3 direction could be a resultant of two or more remanences carried by authigenic minerals magnetized in different directions. The C-4 remanence ($D = 180°$, $I = +10°$) is observed only in dolomitic samples at 3 sites located at the uppermost stratigraphic levels of the Châteauguay sediments.

In view of the lack of adequate faunal control and the predominantly undeformed nature of the Potsdam sediments, it is difficult to determine the relative age of remanences C-1 to C-4 but based on a combination of arguments such as internal consistency of individual remanences and their

frequency of occurrence at sites located in each formation (stratigraphic position), we assigned probable ages of magnetization: C-1 as the earliest remanence acquired in Middle Dresbachian; C-2 is relatively younger (to C-1), some time during Late Dresbachian; C-4 could be Early Ordovician but a very Late Cambrian age cannot be disputed; C-3 may not correspond to a paleomagnetic signal. The corresponding paleomagnetic poles for C-1 and C-2 are located respectively at 46°N, 149°E and 37°N, 156°E (Table 3) which are in reasonably good agreement with other Late Cambrian (Dresbachian to Franconian) poles derived from Texas (Fig. 8). Therefore, a general Late Cambrian age seems indicated for poles C-1 and C-2. This agreement between Cambrian poles derived from Croixian Sauk sea sediments deposited at geographically distant parts (Quebec and Texas) of the North American craton shows the evidence for internal consistency of cratonic poles at least during Late Cambrian time. The younger pole C-4 (40°N, 107°E) is not significantly different from Ordovician and some Late Cambrian poles.

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